



NORTH SYDNEY GIRLS HIGH SCHOOL
YEAR 12 – TERM 1 ASSESSMENT
2006

MATHEMATICS EXTENSION 1

TIME ALLOWED: 60 minutes
Plus 2 minutes reading time

INSTRUCTIONS:

- Start each question on a new page
- Hand each question in separately, including a sheet for non-attempts
- Show all necessary working

This task is worth 20% of the HSC Assessment Mark

Question One (9 Marks)

Marks

- (a) What is the exact value of $\cos\left(\frac{\pi}{6}\right)$? 1
- (b) Differentiate $\cos(x^2 + 1)$ 1
- (c) Find $\int \sec^2 5x \, dx$ 1
- (d) (i) Sketch the curve $y = 4\sin 2x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ 2
- (ii) On your diagram for part (i), sketch the line $y = \frac{1}{3}x$, and shade the region represented by $\int_0^{\frac{\pi}{4}} \left(4\sin 2x - \frac{1}{3}x\right) dx$ 2
- (iii) Find the exact value of the integral in part (ii). 2

Question Two (8 Marks)

- (a) A sphere is being heated so that its surface area is increasing at a constant rate of 25 cm^2 per second. Find the rate of increase of the volume when the radius is 5 cm . 3
- (b) One hundred grams of cane sugar in water are being converted into dextrose at a rate which is proportional to the amount unconverted at any time, i.e. if M grams are converted in t minutes, then $\frac{dM}{dt} = k(100 - M)$, where k is a constant.
- (i) Show that $M = 100 + Ae^{-kt}$, where A is a constant which satisfies the above equation. 2
- (ii) Find the value of A (initially no cane sugar has been converted to dextrose) 1
- (iii) If 40 grams are converted in the first 15 minutes, find how many grams are converted in the first 30 minutes. 2

Question Three (10 Marks)

- (a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin\left(\frac{x}{4}\right)}{3x}$ 2
- (b) (i) Express $\cos x - \sin x$ in the form $R \cos(x + \alpha)$, where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$ 2
- (ii) Hence, or otherwise, solve the equation $\cos x - \sin x = \frac{\sqrt{2}}{2}$ for $0 \leq x \leq 2\pi$ 2
- (c) Prove $\frac{\tan A}{\tan 2A - \tan A} \equiv \cos 2A$ 4

Question Four (10 Marks)

- (a) Find the exact value of $\sin 105^\circ$ 2
- (b) Find the volume of the solid generated when $y = \sin 3x$ is rotated around the x axis from $x = 0$ to $x = \frac{\pi}{3}$. 4
- (c) Differentiate $x \sin 3x$ with respect to x and hence evaluate $\int_0^{\frac{\pi}{2}} x \cos 3x \, dx$ 4

Question Five (9 Marks)

- (a) If $y = \tan 2x$, find the equation of the tangent to the curve at $x = \frac{\pi}{6}$ 3
- (b) Find the acute angle between the lines $4x + y + 5 = 0$ and $6x + 3y - 7 = 0$ correct to the nearest minute. 3
- (c) Solve the equation $\sin 2\theta + \cos \theta = 0$ for $0 \leq \theta \leq 2\pi$ 3

Question Six (10 Marks)

- (a) Solve $5 \sin \theta - 2 \cos \theta = 2$ for $0^\circ \leq \theta \leq 360^\circ$, using the result that $\tan \frac{\theta}{2} = t$ 4
- (b) A particle moves along a straight line so that its displacement, x metres, from a fixed point O is given by $x = 1 - 3 \cos\left(\frac{t}{2}\right)$, where t is measured in seconds.
- (i) Sketch the graph of x as a function of t for $0 \leq t \leq 4\pi$ 2
- (ii) Hence, or otherwise, find when and where the particle first comes to rest after $t = 0$ 2
- (iii) Find a time when the particle reaches its maximum speed. What is this speed? 2

End of paper

Solutions

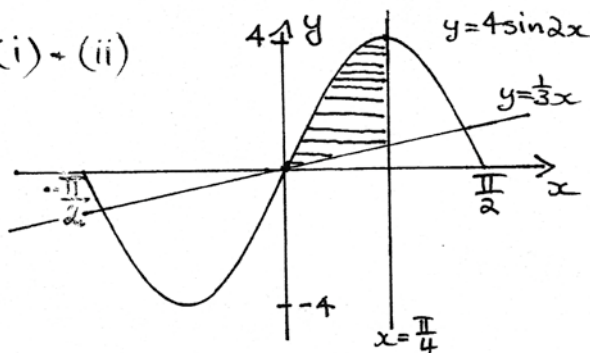
Question one

(a) $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$

(b) $\frac{d}{dx} (\cos(x^2+1))$
 $= -2x \sin(x^2+1)$

(c) $\int \sec^2 5x \, dx = \frac{1}{5} \tan 5x + C$

(d) (i) + (ii)



(iii) $\int_0^{\pi/4} 4 \sin 2x - \frac{1}{3} x \, dx$

$= \left[-2 \cos 2x - \frac{1}{3} \cdot \frac{x^2}{2} \right]_0^{\pi/4}$

$= \left(-2 \cos 2\left(\frac{\pi}{4}\right) - \frac{1}{3} \cdot \frac{\left(\frac{\pi}{4}\right)^2}{2} \right) - \left(-2 \cos 2(0) - \frac{1}{3} \cdot \frac{(0)^2}{2} \right)$

$= \left(-2 \cos \frac{\pi}{2} - \frac{\pi^2}{96} \right) - (-2 - 0)$

$= \left(0 - \frac{\pi^2}{96} \right) - (-2)$

$= 2 - \frac{\pi^2}{96}$

Question two

(a) $\frac{dA}{dt} = 25 \text{ cm}^2/\text{s}$

$A = 4\pi r^2$

$\frac{dA}{dr} = 8\pi r$

$\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}$

$25 = 8\pi r \cdot \frac{dr}{dt}$

$\frac{dr}{dt} = \frac{25}{8\pi r}$

$r = 5, \frac{dr}{dt} = \frac{25}{8\pi \times 5}$
 $= \frac{5}{8\pi}$

$V = \frac{4}{3} \pi r^3$

$\frac{dV}{dr} = 4\pi r^2$

$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$

$= 4\pi r^2 \cdot \frac{dr}{dt}$

$r = 5, \frac{dV}{dt} = 4\pi (5)^2 \times \frac{5}{8\pi}$
 $= \frac{125}{2}$

Volume is increasing by
 $\frac{125}{2} \text{ cm}^3/\text{s}$

Question two

$$(b) (i) \quad M = 100 + A e^{-kt} \\ \Rightarrow A e^{-kt} = M - 100$$

$$\frac{dM}{dt} = -k \cdot A e^{-kt} \\ = -k(M - 100) \\ = -k(100 - M)$$

$$(ii) \quad t=0, \quad M=0$$

$$M = 100 + A e^0$$

$$A = -100$$

$$(iii) \quad M = 100 - 100 e^{-kt}$$

$$t=15, \quad M=40$$

$$40 = 100 - 100 e^{-k \times 15}$$

$$-60 = -100 e^{-k \times 15}$$

$$\frac{3}{5} = e^{-k \times 15}$$

$$\log_e \frac{3}{5} = -k \times 15$$

$$k = -\frac{1}{15} \log_e \frac{3}{5}$$

$$t=30, \quad M=?$$

$$M = 100 - 100 e^{-k \times 30} \\ = 64$$

Question three

$$a) \quad \lim_{x \rightarrow 0} \frac{\sin \frac{x}{4}}{\frac{x}{4}} \times \frac{1}{4}$$

$$= 1 \times \frac{1}{4}$$

$$= \frac{1}{4}$$

Question three

$$(b) \quad \cos x - \sin x \equiv R \cos(x + \alpha) \\ R \cos(x + \alpha) = R(\cos x \cos \alpha - \sin x \sin \alpha) \\ = R \cos x \cos \alpha - R \sin x \sin \alpha$$

$$1 = R \cos \alpha \quad (1)$$

$$-1 = -R \sin \alpha \quad (2)$$

$$\frac{(2)}{(1)} \quad \frac{R \sin \alpha}{R \cos \alpha} = \frac{1}{1}$$

$$\tan \alpha = 1 \\ \alpha = \frac{\pi}{4}$$

$$(1)^2 + (2)^2$$

$$R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 1^2 + 1^2$$

$$R^2 (\cos^2 \alpha + \sin^2 \alpha) = 2$$

$$R^2 = 2$$

$$R = \pm \sqrt{2}, \quad R > 0$$

$$R = \sqrt{2}$$

$$\therefore \cos x - \sin x = \sqrt{2} \cos\left(x + \frac{\pi}{4}\right)$$

$$(ii) \quad \sqrt{2} \cos\left(x + \frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$$

$$\cos\left(x + \frac{\pi}{4}\right) = \frac{1}{2}$$

$$x + \frac{\pi}{4} = \frac{\pi}{3}$$

$$x = \frac{\pi}{12}$$

$$\frac{5\pi}{12}, \frac{7\pi}{12}$$

Question three

$$(c) \text{ LHS} = \frac{\tan A}{\tan 2A - \tan A}$$

$$= \frac{\tan A}{\frac{2\tan A}{1-\tan^2 A} - \tan A}$$

$$= \frac{\tan A}{\frac{2\tan A - \tan A(1-\tan^2 A)}{1-\tan^2 A}}$$

$$= \frac{\tan A}{\frac{2\tan A - \tan A + \tan^3 A}{1-\tan^2 A}}$$

$$= \frac{\tan A}{\frac{\tan A + \tan^3 A}{1-\tan^2 A}}$$

$$= \tan A \times \left(\frac{1-\tan^2 A}{\tan A (1+\tan^2 A)} \right)$$

$$= \frac{1-\tan^2 A}{1+\tan^2 A}$$

$$= \frac{1 - \frac{\sin^2 A}{\cos^2 A}}{1 + \frac{\sin^2 A}{\cos^2 A}}$$

$$= \frac{\frac{\cos^2 A - \sin^2 A}{\cos^2 A}}{\frac{\cos^2 A + \sin^2 A}{\cos^2 A}}$$

$$= \frac{\cos^2 A - \sin^2 A}{1}$$

$$= \cos 2A$$

$$= \text{RHS}$$

Question four

$$\begin{aligned} \text{(a)} \quad \sin 105^\circ &= \sin (60 + 45)^\circ \\ &= \sin 60^\circ \cos 45^\circ + \cos 60^\circ \sin 45^\circ \\ &= \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} + \frac{1}{2} \times \frac{1}{\sqrt{2}} \\ &= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} \\ &= \frac{\sqrt{3}+1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{\sqrt{6}+\sqrt{2}}{4} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad V &= \pi \int_0^{\frac{\pi}{3}} (\sin 3x)^2 dx \\ &= \pi \int_0^{\frac{\pi}{3}} \frac{1}{2} (1 - \cos 6x) dx \\ &= \frac{\pi}{2} \left[x - \frac{1}{6} \sin 6x \right]_0^{\frac{\pi}{3}} \end{aligned}$$

$$\cos 2\theta = 1 - 2\sin^2 \theta$$

$$\text{let } \theta = 3x.$$

$$\cos 6x = 1 - 2\sin^2 3x$$

$$\sin^2 3x = \frac{1}{2}(1 - \cos 6x)$$

$$\begin{aligned} &= \frac{\pi}{2} \left(\left(\frac{\pi}{3} - \frac{1}{6} \sin 2\pi \right) - \left(0 - \frac{1}{6} \sin 0 \right) \right) \\ &= \frac{\pi}{2} \left(\left(\frac{\pi}{3} - 0 \right) - (0) \right) \\ &= \frac{\pi^2}{6} \end{aligned}$$

Volume is $\frac{\pi^2}{6}$ units²

$$\begin{aligned} \text{(c)} \quad \frac{d}{dx} (x \sin 3x) \\ &= x \times 3 \cos 3x + \sin 3x \times 1 \\ &= 3x \cos 3x + \sin 3x \end{aligned}$$

$$\int_0^{\frac{\pi}{2}} 3x \cos 3x dx = \int_0^{\frac{\pi}{2}} \frac{d}{dx} (x \sin 3x) - \sin 3x dx.$$

$$\begin{aligned} \int_0^{\frac{\pi}{2}} x \cos 3x dx &= \frac{1}{3} \left(x \sin 3x + \frac{1}{3} \cos 3x \right) \Big|_0^{\frac{\pi}{2}} \\ &= \frac{1}{3} \left(\left(\frac{\pi}{2} \sin \frac{3\pi}{2} + \frac{1}{3} \cos \frac{3\pi}{2} \right) - \left(0 + \frac{1}{3} \cos 0 \right) \right) \\ &= \frac{1}{3} \left(\left(-\frac{\pi}{2} + 0 \right) - \left(\frac{1}{3} \right) \right) \\ &= -\frac{\pi}{6} - \frac{1}{9} \end{aligned}$$

Question five

(a) $y = \tan 2x.$

$$\frac{dy}{dx} = 2 \sec^2 2x.$$

$$x = \frac{\pi}{6}, \quad \frac{dy}{dx} = 2 \sec^2 \frac{\pi}{3} \\ = 2 \times \frac{1}{(\frac{1}{2})^2}$$

$$= 8$$

$$y - \sqrt{3} = 8(x - \frac{\pi}{6}) \\ y - \sqrt{3} = 8x - \frac{4\pi}{3}.$$

$$y = \tan \frac{\pi}{3} \\ = \sqrt{3}.$$

(b) $y = -4x - 5$
 $m_1 = -4$

$$3y = -6x + 7 \\ m_2 = -2$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ = \left| \frac{-4 - (-2)}{1 + (-4)(-2)} \right|$$

$$\tan \theta = \left| \frac{-2}{9} \right|$$

$$\theta = 12^\circ 32'$$

(c) $\sin 2\theta + \cos \theta = 0$

$$2\sin \theta \cos \theta + \cos \theta = 0.$$

$$\cos \theta (2\sin \theta + 1) = 0$$

$$\cos \theta = 0$$

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$2\sin \theta = -1$$

$$\sin \theta = -\frac{1}{2}$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}.$$

Question six

$$(a) \quad 5 \sin \theta - 2 \cos \theta = 2$$
$$5 \left(\frac{2t}{1+t^2} \right) - 2 \left(\frac{1-t^2}{1+t^2} \right) = 2.$$

$$5(2t) - 2(1-t^2) = 2(1+t^2)$$
$$10t - 2 + 2t^2 = 2 + 2t^2$$

$$10t = 4$$

$$t = \frac{2}{5}$$

$$\tan \frac{\theta}{2} = \frac{2}{5}$$

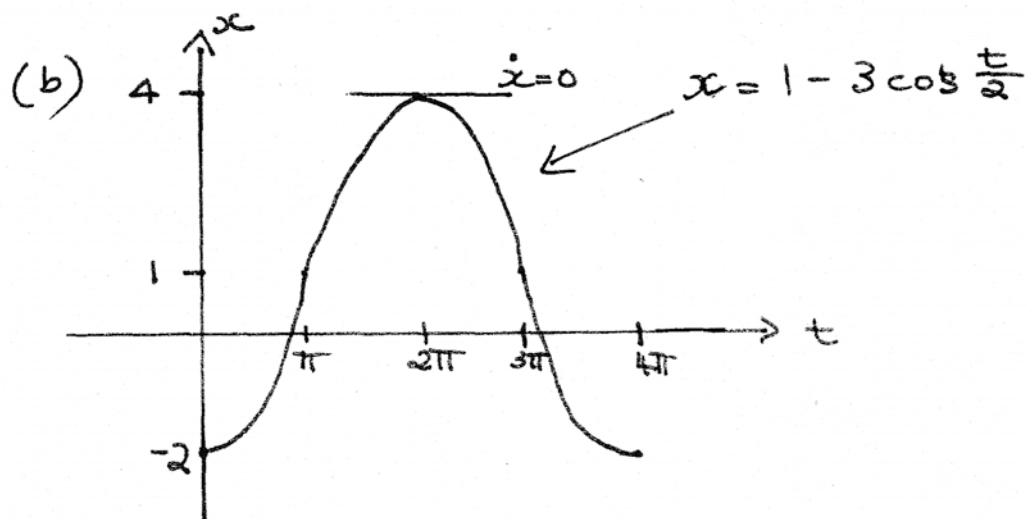
$$\left(\frac{\theta}{2} = 21^\circ 48' \right)$$

$$\theta = 43^\circ 36'$$

Check 180°

$$5 \sin 180^\circ - 2 \cos 180^\circ = 0 - 2 \times -1$$
$$= 2$$
$$= \text{RHS.}$$

$$\therefore \text{Soln } \theta = 43^\circ 36', 180^\circ$$



$$(ii) \quad \dot{x} = 0, \quad t = 2\pi \text{ s} \quad x = 4 \text{ m}$$

$$(iii) \quad \text{max speed } t = \pi \text{ s (or } 3\pi)$$

$$\dot{x} = \frac{3}{2} \sin \frac{t}{2} = \frac{3}{2} \sin \frac{\pi}{2} = \frac{3}{2}$$

$$\therefore \text{max speed} = \frac{3}{2} \text{ m/s}$$